4a. INTRODUCTION TO ROBOT PROGRAMMING

AIM

To get introduced to Robots, their types, terminology and programming methods.

INDUSTRIAL ROBOT

The RIA (Robotic Industries Association) defines the robot in the following way "An industrial robot is a programmable, multifunctional manipulator designed to move materials, parts, tools or special devices through variable programmed motion for the performance of a variety of robots.

THREE LAWS OF ROBOTICS

- A robot must not injure any human.
- A robot must obey the order given by the human unless it conflicts with the first law.
- A robot must protect itself and its existence unless it conflicts with the previous laws.

ROBOT TERMINOLOGY

- Robotics - The study of robots.
- DOF (Degrees Of Freedom) - The number of independent motions a device can make.
- Manipulator - A device capable of integrating with the environment.
- Anthropomorphic - Like human being.
- End Effectors - The tool, gripper or any other device mounted at the end of manipulator for accomplishing useful task
- Work Space - The volume in space where a robot end effector can reach both in position and orientation.
- Link - A rigid piece of material connecting joints in a robot.
- Joint - A device which allows relative motion between links of a robot.
- Dynamics - The study of motion under forces.
- Load Bearing Capacity - The maximum weight carrying capacity of a robot.

FLEX PENDANT

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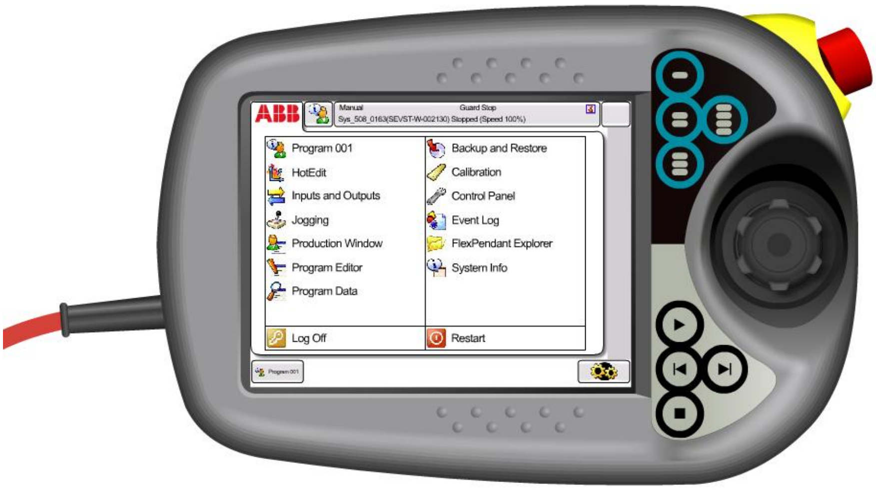
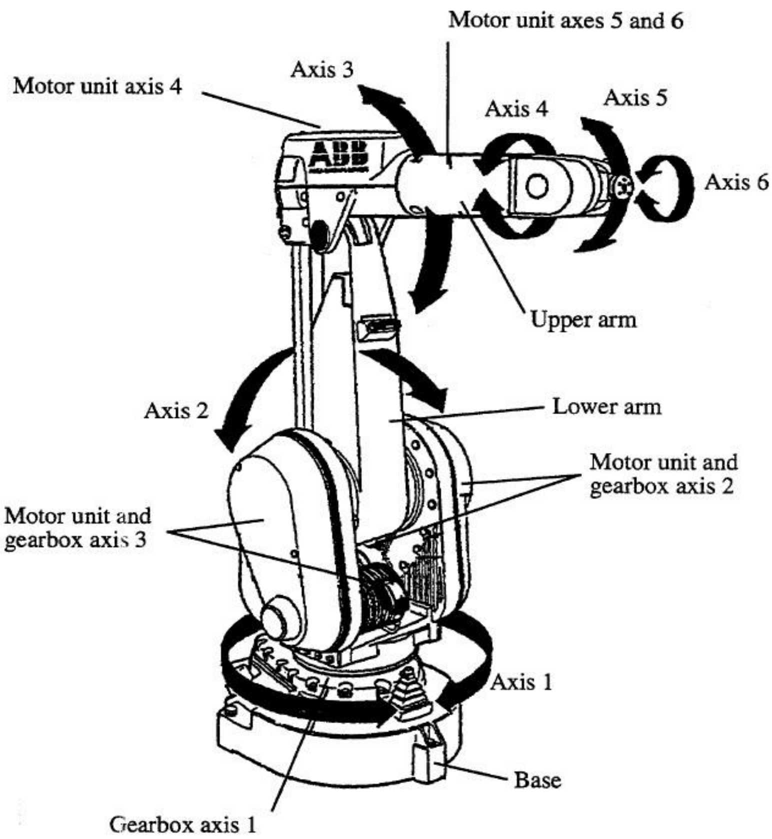


ABB 6 - AXIS ROBOT



TYPES OF ROBOT JOINTS

• Rotational	-	This enables the robot to place its arm in any direction on a horizontal plane.
• Radial	-	This enables the robot to move its end effector radially to reach distant points.
• Vertical	-	This enables the robot to take its end effector to different heights.

SIX AXIS OF ROBOT

The first three axis are for positioning and the remaining three axis are for orientation of the end effector.

DESCRIPTION OF THE CONTROLLER

The controller is the brain behind the functioning of robot. The latest one is IRC5. The major components of it are:

- A main computer does all the primary computing.
- Axis computer performs all the calculations related to joints.
- Drive unit control the torque acceleration and speed of the joints.
- The SMPS supplies 24v DC to main computer, axis and input ports.
- The capacitor improves the power factor.
- Contactors cut - off supply from motor as per requirements.
- Transformers steps down 440v AC to 230v AC.
- Rectifier converts AC to DC.
- Input boards are used for user signals.

BASIC ROBOT PROGRAMMING

- The language used by ABB is rapid.
- Program can be edited by program editor.
- Select new for writing a new program.
- Soft is used for writing a program.

MOTION INSTRUCTIONS

1. MOVE J, p1, v500, z250, tool 0;
2. MOVE L, p1, v500, z250, tool 0;
3. MOVE C, p1, p2 ,v500, z250, tool 0;
4. MOVE AbsJ ,p1, v500,z250,tool 0;

Here p1 and p2 are user defined points.

MOVE L	**:**	Moves the TCP linearly
MOVE C	**:**	Moves the TCP circularly
MOVE J	**:**	Moves the robot by joint movement
MOVE AbsJ	**:**	Moves the robot to an absolute joint position

Hand Drawn

COORDINATE SYSTEMS

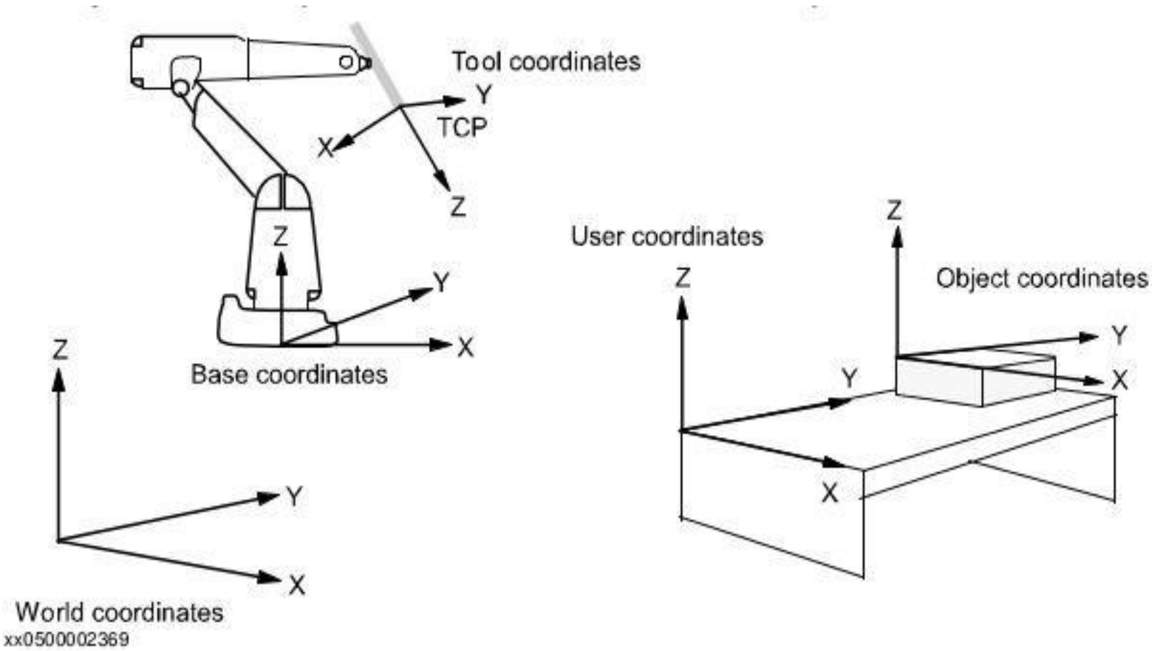


ABB - IRC5 CONTROL SYSTEM WITH CONTROL UNIT FOR SCC

ITEM	DESCRIPTION
1	Main power switch
2	Emergency stop
3	Motors ON button
4	Operating mode selector
5	Diode panel that shows the status of the safety loops (optional)
6	Running time meter

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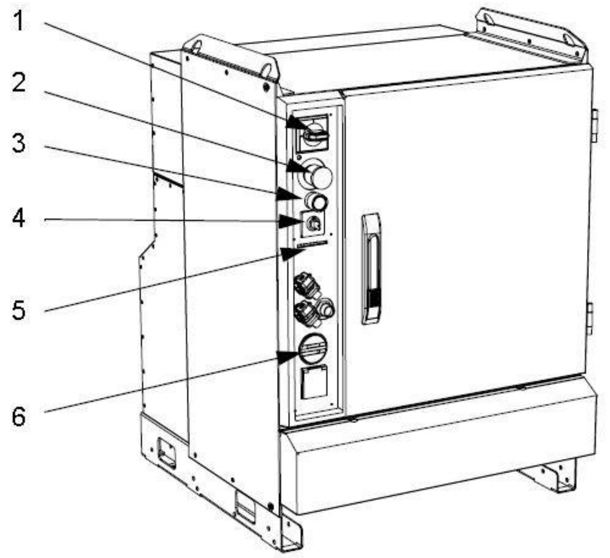
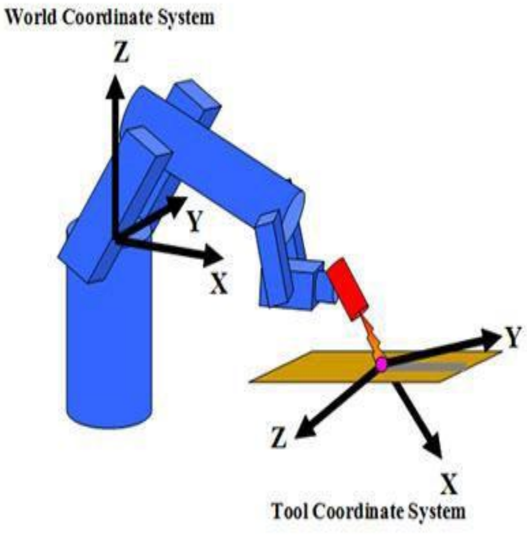
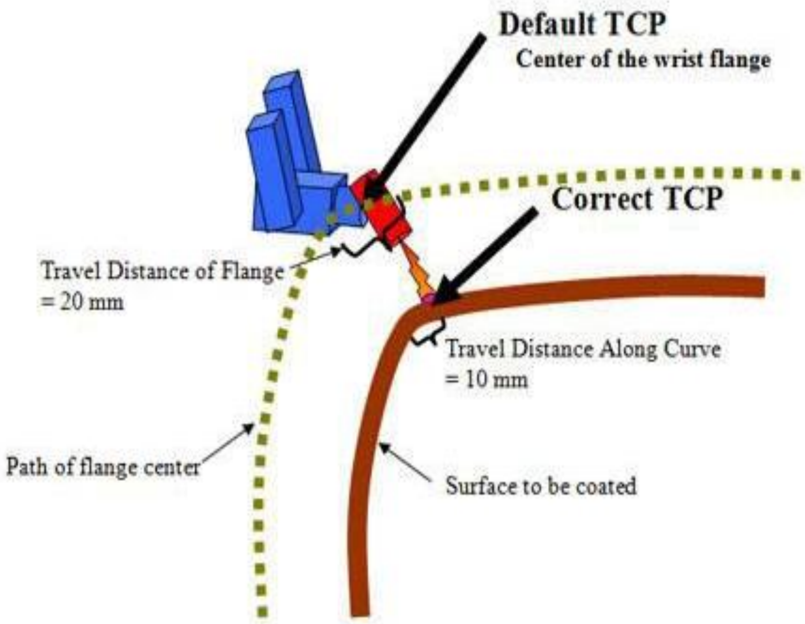


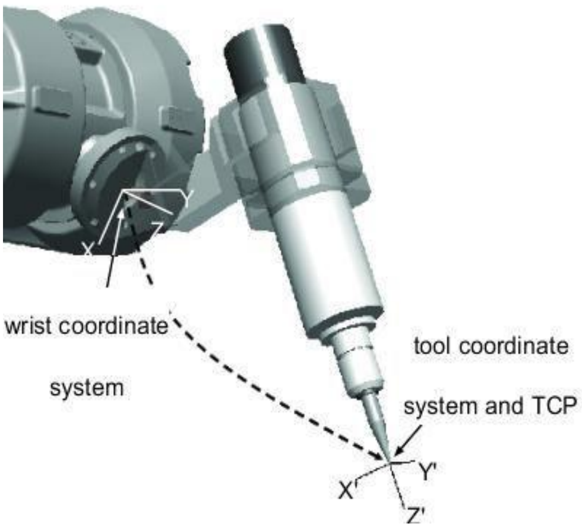
TOOL COORDINATE SYSTEM FOR ARTICULATED ROBOT



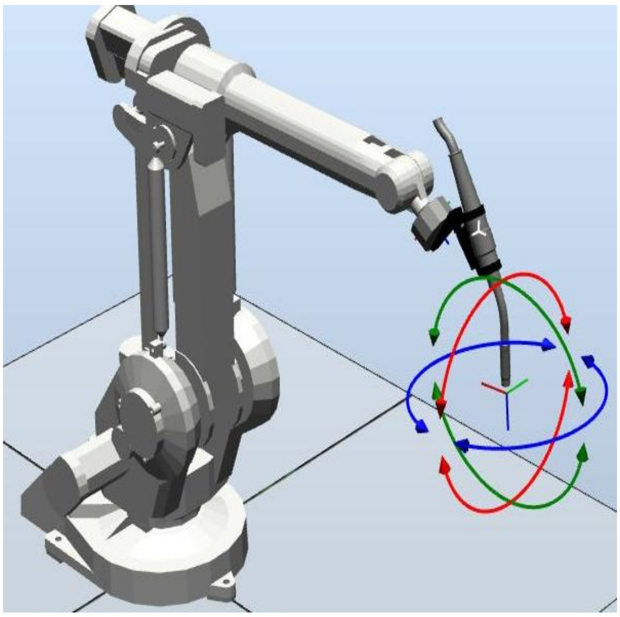
ROBOT MOTION OF TCP



WCS



TCS



RESULT

The basics of robot programming has been studied successfully using IRB1410 6 axis serial manipulator

4 b - PICK AND PLACING THE OBJECTS USING JOGGING (IRB1410)

AIM

To do the pick and place operation using the manual mode of the ABB robot.

EQUIPMENT REQUIRED

- ABB ROBOT
- Peg along holders
- Gripper with its supply

PROCEDURE

- Give the pre - electrical supplies to the robot.
- White light will start blinking.
- Grip the flex pendant appropriate and press the end
- Select jogging mode.
- Before doing the above, check if manual mode is selected.
- Axis, Linear and re - oriental modes need to be used appropriately along with parameters such as speed is adjusted as per need.
- The gripper must be opened before the exact position of 80% gripping.
- Go to ABB main menu and go to Input and Output.
- Set d01 and reset d03 to close the gripper.
- The gripper grips the block and slowly lifts its arm.
- To open the grippers go to Input and Output and reset d01 and sent d03.
- Move the arm to a safe position after the object is dropped at dropped at the drop point.

 # Safety: for safer operation don't stand with in the work volume of the robot and while running the program / Executing the task

RESULT

Thus, the pick and place operation of a wooden peg from one holder to another was performed successfully